Transmittal Email

To: Verna Johnson, CEO of BTB Heavy Equipment, Inc.

From: Skylar Magouliotis, Team Lead at Core Systems

Date: 2026-04-20

Subject: Core Systems - Response to Request for Proposal (03-2026a)

Dear Mrs. Johnson,

Our team at Core Systems received a Request for Proposal (03-2026a) from BTB Heavy Equipment, Inc. We are interested in working with you on this project!

The team at Core Systems is highly skilled and possess all the certifications that are required for this project. Core Systems is a local Saskatchewan-based IT company. We provide systems and machines, protect data, and support businesses throughout Saskatchewan.

In response to BTB Heavy Equipment's Request for Proposal (03-2026a), we would like to provide you with a solution for the standard desktop computers that you require for approximately 250 end-users by June 2026. After thorough research, we have determined the best components for these standard desktop computers, performed cost analysis to ensure that you do not need to utilize the full budget, offer data security and installation support, and give recommendations on disposal methods and other considerations.

Core Systems is passionate about supporting the local businesses in our community, and we would love to support BTB Heavy Equipment with their mission to offer contracts to the most reliable brands with exceptional service.

Our team at Core Systems will provide a research-based practical solution to BTB Heavy Equipment's Request for Proposal that will enhance efficiency, back up important data, consider sustainability, and support your company strategy.

We look forward to working with you!

Sincerely,

Skylar Magouliotis

Team Lead
Core Systems

Driving Operational Efficiency: A Comprehensive Hardware Infrastructure Upgrade

RFP Response for BTB Heavy Equipment, Inc

Request for Proposal (03-2026a)

BTB Heavy Equipment, Inc

2026-04-16



Core Systems

Connect, Implement, Thrive

Skylar Magouliotis

Savneet Kaur

Matthew Livingstone

Lucas McCulloch

Land Acknowledgement

Core Systems would like to acknowledge that we are on treaty land, referred to as Treaty 4 Territory in Regina. Treaty 4 encompasses the lands of the Cree, Saulteaux, Dakota, Nakota, Lakota, and on the homeland of the Métis Nation.

Executive Summary

This proposal will offer a solution to the Request for Proposal (03-2026a) put forth by BTB Heavy Equipment, Inc.

BTB Heavy Equipment requires updated standard desktop computers to support 250 end users by June of 2026. We will need to ship the components as soon as possible to ensure this project does not exceed the timeline as stated in Appendix B. Each standard desktop computer will be created from individual components. In this document, we list each component in the standard desktop computer, the associated specifications and requirements for the component, describe the function, and provide a reason for selection. A central processing unit (CPU) will allow BTB Heavy Equipment to run Windows 11 Professional and will perform well during basic computing tasks. The motherboard is compatible with all the selected components, will have room for future memory upgrades, and will allow you to manage the device easily in a business environment. The memory (8GB) will hold data and is the most cost-effective option that meets the requirements for your company. A solid-state drive (SSD) will act as storage for the standard desktop computer; it exceeds the size requirement, and it is faster than traditional hard disks. A network interface card (NIC) connects the computer to the network and allows the company to operate on a Gigabit Ethernet LAN and use the internet. A power supply provides the necessary power to the standard desktop computer; this unit is energy efficient and has more wattage capacity than what was required for this build. A card reader with CompactFlash™ support will allow the company to read, open, and transfer data using a CompactFlash™ card that is compatible with Windows 11 Professional. The motherboard did not contain FireWire (which is an IEEE 1394 port), so it was included as an expansion card and supports fast data transfer. An optical drive allows your company to transfer data and burn disks. A 22-inch Full HD monitor and other peripherals will allow the user to utilize the standard desktop computer efficiently. All the parts that we selected for the standard desktop computer ensure complete system functionality.

Based on the cost analysis for each component, we estimate that 250 standard desktop computers and the associated environmental handling fees with be $234,082.50 of the $250,000.00 budget as listed in the budget on page 19. This will save BTB Heavy Equipment approximately between $15,917.50 - $28,000.00 depending on sale cost(s) at the time of shipping.

Once the standard desktop computers are built, we will back up important data from BTB Heavy Equipment's current machines, and the dispose of the machines in an ethical and sustainable way through proper recycling methods and re-use in community projects listed in Appendix C – Disposal Solutions. The skilled technicians at Core Systems will supervise every installation to ensure that BTB Heavy Equipment has standard desktop computers that are able to run Windows 11 Professional with Office 365 on a Gigabit Ethernet LAN.

Abstract (Technical Summary)

This proposal will offer a solution to the Request for Proposal (03-2026a) put forth by BTB Heavy Equipment, Inc.

Once selected, Core Systems will immediately ship the selected components, build 250 new standard desktop computers, and deliver them promptly. We will ensure that the standard desktop computers will be available to approximately 250 end users in June of 2026 (as described in Appendix B – Proposed Project Timeline).

Our team researched and performed cost analysis on the parts that we selected for the standard desktop computers. The Intel Core i3-8300 processor with integrated graphics will allow BTB Heavy Equipment to run Windows 11 Professional and will perform well during basic computing tasks. The ASUS H110M-K motherboard with integrated sound allows the CPU to work with the socket type and has dual-channel architecture for memory updates, as well as WOL by PME which allows you to manage the device efficiently in a business environment. 8GB DDR4 ECC memory holds data and is the most affordable option for the company's requirements. A 512GB SATA III solid-state drive will act as storage for the standard desktop computer; it exceeds the size requirement, and it is faster than traditional hard disks. A 2.5Gbps network interface card connects the computer to the network and allows the company to operate on a Gigabit Ethernet LAN. The Montech Century II 850 power supply is energy efficient and has more wattage capacity than required for this build. A SmartQ 5Gbps card reader with CompactFlash™ support will allow the company to read, open, and transfer data and use a CompactFlash™ card that is compatible with Windows 11 Professional. The FireWire card is an IEEE 1394 port that was not integrated into the motherboard, so it was included as an expansion card and supports fast data transfer. The ORIGBELIE BT669 5Gbps optical drive is an external drive that will allow the company to transfer data and burn disks. A 22-inch Full HD monitor and other peripherals will allow the user to utilize the standard desktop computer efficiently. All the parts that we selected for the standard desktop computer ensure complete system functionality.

The cost of this project is estimated to be $234,082.50 of the $250,000.00 budget, which will save BTB Heavy Equipment approximately between $15,917.50 - $28,000.00 depending on sale cost(s) at the time of shipping.

Our skilled technicians (as listed in Appendix G) will back up important data from BTB Heavy Equipment's current machines and then dispose of the machines ethically and sustainably through proper recycling methods and re-use with community organizations (listed in Appendix C – Disposal Solutions).

Every standard desktop computer will be able to run Windows 11 Professional with Office 365 and will operate efficiently on a Gigabit Ethernet LAN.

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Formal Introduction

This proposal will offer a solution to the Request for Proposal (03-2026a) put forth by BTB Heavy Equipment, Inc. The objective of this proposal is to convey the results of the research conducted by Core Systems. Our team at Core Systems would like to provide the end-users at BTB Heavy Equipment with the best standard desktop computers for the best possible cost by June 2026.

This document will provide Verna Johnson (CEO of BTB Heavy Equipment), stakeholders, executives, technicians, and the IT department, with details regarding the individual components for each standard desktop computer as stated in the project description; this description will include the associated specifications and requirements for the component, the function, and the reason for selection by our team. User reviews for selected components will be listed in Appendix F. The standard desktop computers that we have selected will allow your company to execute administrative tasks, data analytics, inventory management, enhanced communication, project management, machine loaning, and other objectives. Every standard desktop computer is designed to run Windows 11 Professional with Office 365 on a Gigabit Ethernet LAN. We thoroughly researched every part that was necessary to include based on the requirements that BTB Heavy Equipment indicated in the Request for Proposal (03-2026a). It was imperative for Core Systems to provide the standard desktop computers that possess all the necessary requirements, so that BTB Heavy Equipment can continue to offer contracts for the most reliable brands with exceptional customer service and support Saskatchewan businesses.

Technical terminology that is stated in this document will be described in Appendix A - Glossary.

We performed cost analysis on every component and determined that this project is estimated to be $234,082.50 of the $250,000.00 budget, which will save BTB Heavy Equipment approximately between $15,917.50 - $28,000.00 depending on sale cost(s) at the time of shipping. This cost analysis includes environmental handling fees, but does not include taxes, labour estimates, or other associated costs due to budget limitations. The budget can be found on page 19 of this document. Please note that keeping any current functional peripherals and omitting the FireWire component will make the final total cost decrease. This is not necessary, but it is what Core Systems would recommend if BTB Heavy Equipment is interested in saving approximately $5000 in additional savings. This requirement is explained further in the project description under the FireWire component on page 7. While researching components, our team determined that there was an extreme increase in the cost of RAM recently; this was not an issue as we were able to find an 8GB RAM module that would allow the company to utilize the system without any additional RAM, but we decided to select a motherboard that had dual-channel architecture to allow

more RAM to be installed if the company would like additional RAM modules. BTB Heavy Equipment may want to consider this in the future, but it depends on what you are planning on doing with the standard desktop computers in upcoming years. RAM considerations are outlined in Appendix E.

Once selected, Core Systems will ship the components, build the standard desktop computers, and determine what data needs to be transferred or backed up from the current machines. We specialize in data transfer and security, so we will do this prior to installation. This can take time, so we will need to start this process as soon as possible as noted in Appendix B – Proposed Project Timeline. Then, we will dispose of the current machines based on BTB Heavy Equipment's preferred method as stated in Appendix C – Disposal Solutions.

Core Systems will supervise every installation to ensure that BTB Heavy Equipment has standard desktop computers that are able to run Windows 11 Professional with Office 365, and then we will test each machine on-site to ensure that they operate efficiently on a Gigabit Ethernet LAN. We are certified and highly skilled technicians (as listed in Appendix G), and we look forward to working as a team with your technicians at BTB Heavy Equipment to ensure a fast and error-free deployment. We expect the whole operation from shipping to deployment to take approximately 9 weeks.

Our team at Core Systems will provide a research-based practical solution to BTB Heavy Equipment's Request for Proposal that will enhance efficiency, back up important data, consider sustainability, and support your company strategy.

Project Description

Introduction

BTB Heavy Equipment will require 250 new standard desktop computers that are able to run Windows 11 Professional with Office 365 on a Gigabit Ethernet LAN. Each standard desktop computer will be created from individual components. This document will consist of main components, secondary components, and peripherals. Core Systems will make the recommended selection of each part based on compatible features and hardware. In this document, we will list each part that is a component of the standard desktop computer, the associated specifications and requirements for the part, describe the function, and provide a reason for selection. Specific definitions and terminology can be found in Appendix A – Glossary.

Our team thoroughly researched each individual part that we selected for the standard desktop computers. The Intel Core i3-8300 processor with integrated graphics will allow BTB Heavy Equipment to run Windows 11 Professional and will perform well during basic computing tasks. The ASUS H110M-K motherboard with integrated sound allows the CPU to work with the socket type and has dual-channel architecture for memory updates, as well as WOL by PME which allows you to manage the device efficiently in a business environment. 8GB DDR4 ECC memory holds data and is the most affordable option for the company's requirements. A 512GB SATA III solid-state drive will act as storage for the standard desktop computer; it exceeds the size requirement, and it is faster than traditional hard disks. A 2.5Gbps network interface card connects the computer to the network and allows the company to operate on a Gigabit Ethernet LAN. The Montech Century II 850 power supply is energy efficient and has more wattage capacity than required for this build. A SmartQ 5Gbps card reader with CompactFlash™ support will allow the company to read, open, and transfer data and use a CompactFlash™ card that is compatible with Windows 11 Professional. The FireWire card is an IEEE 1394 port that was not integrated into the motherboard, so it was included as an expansion card and supports fast data transfer. The ORIGBELIE BT669 5Gbps optical drive is an external drive that will allow the company to transfer data and burn disks. A 22-inch Full HD monitor and other peripherals will allow the user to utilize the standard desktop computer efficiently. All the parts that we selected ensure complete system functionality.

Every requirement is met within budget, and all components are integrated or compatible with one-another. Thorough research and cost analysis was performed on every component; reviews, specifications, and requirements were considered during this analysis to ensure that BTB Heavy Equipment will have the best standard desktop computers for their end users. Our research-based selection of parts will enhance efficiency, back up important data, consider sustainability, and support your company strategy.

Central Processing Unit (CPU)

The central processing unit (CPU) acts as the brain of the computer and it is responsible for the flow of data (Andrews, Shelton, & Pierce, 2023). The CPU is also considered a microprocessor or processor. This component receives data input, processes information, and executes instructions (Andrews, Shelton, & Pierce, 2023). Nearly all the processing of data and instructions will occur in the central processing unit.



Figure 1 – Intel i-3-8300 Central Processing Unit

This CPU is an 8th generation processor and can support the client requirement that states that the machine must be able to run Windows 11 Professional with Office 365 on a Gigabit Ethernet LAN (King & Rayome, 2025). This CPU has 4 cores, can support up to 3 monitors, has a built-in fan/heat sink, and is compatible with the motherboard (Amazon, 2026). This CPU has integrated graphics (Intel UHD Graphics 630), and the integrated part can handle basic computing tasks, and HD video playback without requiring a separate graphics card (Amazon, 2026). The specifications for the integrated video card are located on page 5 of this document.

Table 1 – Specifications and Requirements for the Intel i-3-8300 Central Processing Unit

Specifications	
Manufacturer	Intel
Slot/Socket Type	LGA-1151
Model	i3-8300
Speed	3.7GHz
Requirements	
Dual Core	☑
At least 2.5GHz	☑
Matches motherboard slot/socket type	☑

Video Card

The video card is also known as a graphics processing unit (GPU) and it is responsible for rendering and displaying images, videos, and animations on a monitor (Lenovo, 2026).This video card can support 4k at 60Hz and ECC Memory (Amazon, 2026) which helps with data integrity (Andrews, Shelton, & Pierce, 2023). This video card is already integrated into the CPU, so it saves space inside of the case and can still perform the computing tasks required for this build.

We chose this part because it is already onboard (integrated), which saves time on installation, and it can handle all the basic computing tasks that the end users are responsible for. "Most motherboards sold today have one or more video ports integrated into the motherboard and are called onboard ports" (Andrews, Shelton, & Pierce, 2023) which will eliminate the need for an additional video card.

Table 2 – Specifications and Requirements for the Intel UHD Graphics 630

Specifications	
Manufacturer	Intel
Model (Version)	31.0.101.2141
Chip Set	UHD Graphics 630
Video Memory	64 GB
Maximum resolution supported	HDMI: 4096x2304@24Hz DP & eDP – (integrated flat panel): 4096x2304@60Hz
Special Features	Supports 4k at 60Hz Supports ECC Memory
Requirements	
PCI Express	☑
AGP Card	N/A (Integrated)
VGA Port	N/A (Integrated)
No video inputs	☑

Motherboard

The main board in the computer is the motherboard. Nearly everything is plugged into the motherboard, and it acts as a hub for other components to communicate. The motherboard can also be referred to as the system board (Andrews, Shelton, & Pierce, 2023). Components that can be plugged into the motherboard include (but are not limited to) the CPU, ROM chips, DIMMs, and interface cards (Andrews, Shelton, & Pierce, 2023).



Figure 2 – ASUS Micro-ATX Motherboard

We selected this motherboard because it supports integrated components such as video and sound, it meets the requirements for the CPU to work with the socket type, and it has features like dual-channel architecture. WOL by PME is present because our NIC is not integrated (newegg, 2023). BTB Heavy Equipment requested that the machines run Windows 11 Professional with office 365 on a Gigabit Ethernet LAN; this motherboard has a Gigabit Lan Controller and LAN (RJ45) port (newegg, 2023). Dual-channeling architecture allows more RAM to be installed if the company wishes to increase the number of memory modules, and they may want to consider this due to the Microsoft 11 requirements.

Table 3 – Specifications and Requirements for the ASUS Micro-ATX Motherboard

Specifications	
Manufacturer	ASUS
Form Factor	Micro ATX
Bus Speed	DDR4 2400 MHz
Slot/Socket Type	LGA-1151
Chip Set	INTEL H110
Onboard Connectors	2 onboard USB ports, 2 additional USB ports & 1 DVI port (there are more not listed here)
Integrated Components	Supports integrated video & sound
Requirements	
Correct slot/socket type for CPU	☑
Correct slot for memory type and speed	☑
Dual-channel architecture	☑
Minimum 4 USB ports	☑
IEEE 1394 port	Not integrated – Expansion Card
WOL if NIC not integrated	☑

FireWire Expansion Card (IEEE 1394 Port)

FireWire, also known as IEEE 1394, is a high-speed interface technology commonly used for connecting external devices to computers (Sharrow, 2025). FireWire, known for its high-speed data transfer capabilities and widespread use in various devices, plays a crucial role in numerous industries (Sharrow, 2025). Users needing FireWire connectivity may need to explore alternative solutions like USB-C adapters or external FireWire to USB converters to ensure compatibility with Windows 11 (Sharrow, 2025). The motherboard component does not have an IEEE 1394 (FireWire) port, so we selected an expansion card that can provide this requirement.



Figure 3 – axGear IEEE-PCI-E FireWire Expansion Card

Windows 11 has more security regarding driver requirements, and this can cause compatibility issues for FireWire devices (Sharrow, 2025). Given the increasing focus on USB and Thunderbolt connectivity, FireWire may face challenges in maintaining its relevance, and the future of FireWire in Windows 11 ultimately depends on whether it can adapt to the current market (Sharrow, 2025). We included it here and in the budget; however, FireWire may not be necessary, and it may be in the company's best interest to omit this requirement (to save money, time, and still obtain functional and efficient machines that meet all the other requirements).

We chose this expansion card because it is compliant with PCI Express Base Specification 1.0a, and PCI Express 1-lane(x1) FireWire adapter will work with PCI Express slots with different lane width (Amazon, 2026). This card supports data transfer rates up to 400Mbps and can support DV camcorders, hard disk, removable drives, scanner, digital cameras, and other FireWire audio/video devices (Amazon, 2026). Other users with Windows 11 did not have any issues with this specific card (see Appendix F for user reviews).

Table 4 – Specifications and Requirements for the axGear IEEE-PCI-E FireWire Expansion Card

Specifications	
Brand	axGear
Hardware Interface	IEEE 1394 (FireWire), PCI
Model Number	IEEE-PCI-E
Connectors	2 x 6 Pin Connection 1 x 4 Pin Connection
Requirements	
Compatible with Motherboard	☑

Memory (RAM)

Random access memory (RAM) stores data to be accessed by the CPU. When the computer is turned off, information in RAM is lost. RAM is a module on the motherboard that contain microchips used to temporarily hold data and programs while the CPU processes both (Andrews, Shelton, & Pierce, 2023).



Figure 4 – Micron 8GB RAM Module

This random-access memory (RAM) was the best option because of its ECC capabilities, compatibility with the motherboard, and price. RAM prices are exponentially higher right now (as noted in Appendix E) compared to previous years and this was an affordable option that could meet BTB Heavy Equipment's requirement for Windows 11 Professional (with the module size being Core Systems recommended minimum size for this specific requirement). The client can always add more RAM, but RAM prices have gone up significantly this year, and adding additional RAM could cost exponentially more (as noted in Appendix E – RAM Considerations).

Table 5 – Specifications and Requirements for the Micron 8GB RAM Module

Specifications	
Manufacturer	Micron
Type	DIMM
Module size	8GB
Speed	2400 MHz
Other Properties	1.2 volts, DDR4, ECC Capable, registered (buffered)
Requirements	
Minimum 4 GB	☑
Single DIMM Preferred	☑
Parity/ECC Capabilities	☑

Network Interface Card (NIC)

Network interface cards (NIC) are also called network adapters. The NIC is an expansion card that is inserted into the motherboard and provides a port to connect the computer to a network (Andrews, Shelton, & Pierce, 2023). Expansion cards are frequently used to enhance the capabilities of the computer (Andrews, Shelton, & Pierce, 2023). The part is essential for this build because it provides a reliable communication channel through an ethernet cable (Lenovo, 2026).

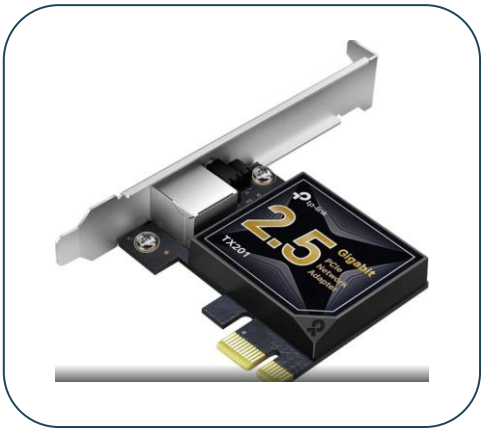


Figure 5 – TP-Link TX201 NIC

Core Systems chose this part for the NIC because it is 2.5x faster than standard Gigabit Ethernet and will be ideal for the bandwidth-intensive tasks such as local data transfer and Internet access (newegg, 2026). This NIC is a wired PCIe card so it has more secure data transmission, has low latency, has consistent performance regardless of distance with cable limits, and it provides a highly stable and reliable network connection (GeeksforGeeks, Sanchhaya Education Private Limited, 2025). Wireless architecture, in comparison, does not perform to the required standard that we want to implement. This will provide your company with the ability to run the standard desktop computers on a Gigabit Ethernet LAN.

Table 6 – Specifications and Requirements for the TP-Link TX201 NIC

Specifications	
Manufacturer	TP-Link
Model	TX201
Architecture	Wired PCI Express Interface Card
Speed	2.5Gbps
Special Features	Faster than standard Gigabit Ethernet Fast local data transmissions Backwards compatible with multiple data rates Compatible with Windows 11
Requirements	
PCI card	☑
Gigabit Ethernet with WOL	☑

Hard Drive (SSD)

The SSD (solid-state drive) acts as storage for the computer. It is an electronic device that does not have any moving parts and uses memory chips to store data instead of spinning disks (Andrews, Shelton, & Pierce, 2023). Some examples include jump drives (also known as thumb drives), flash memory cards, and solidstate disks (Andrews, Shelton, & Pierce, 2023).



Figure 6 – Team Group Vulcan Z SSD

This solid-state drive (SSD) is the ideal drive for this build because it has over 500GB of storage and is faster than traditional hard disk drives (HDDs) (newegg, 2022). There are no RPM's listed because this is an SSD, not an HDD; instead, the performance is determined by read and write times. The maximum sequential read is up to 540 MBps, and the maximum sequential write is up to 470 MBps, where the TBW (terabytes written) is 120TB, and the MTBF (mean time between failures) is 1,000,000 hours (newegg, 2022). This means that BTB Heavy Equipment will be able to utilize this component for a long period of time and the performance is reliable.

Table 7 – Specifications and Requirements for the Team Group Vulcan Z SSD

Specifications	
Manufacturer	Team Group
Model	Vulcan Z SSD (T253TZ512G0C101)
Size	512GB
RPM's	N/A for SSD
ATA Standard	SATA III
Requirements	
Minimum 500GB	☑
Attaches to standard IDE or SATA cable	☑

Power Supply

The power supply supplies power to the machine and peripherals. The power supply can also be referred to as the power supply unit (PSU) and is a box inside of the computer enclosure (Andrews, Shelton, & Pierce, 2023). The power supply will receive power and convert it for use by the motherboard and other installed devices (Andrews, Shelton, & Pierce, 2023).



Figure 7 – Montech Century II 850 Power Supply

This power supply is energy efficient (80 plus gold certified) and has a maximum power output of 850W which is ideal for the machine to run systematically (newegg, 2025). We needed a power supply with this watt rating because it needs to be able to effectively power all the parts that we are putting in the machine, and this part has more wattage capacity than required for this build. A full modular power supply was selected because it provides a detachable cable system, reduces clutter, and improves air flow in the enclosure (newegg, 2025). This power supply is compatible with the latest platforms, is Cybenetics Platinum Certified[1], has reliable components, has optimized ventilation with a 135mm FBD silent fan with zero RPM mode, and has equipped OCP, OVP, UVP, SCP, OTP, and OPP (newegg, 2025). Full definitions for the terms OCP, OVP, UVP, SCP, OTP, and OPP can be found in Appendix A – Glossary.

Table 8 – Specifications and Requirements for the Montech Century II 850 Power Supply

Specifications	
Manufacturer	Montech
Model	Century II 850
Watt Rating	850 (80 plus gold certified)
Requirements	
350W or higher	☑
Matches motherboard connector type	☑

[1] ≥89% & <91% overall efficiency; ≥0.975 Power Factor; >76% 5VSB Efficiency; <0.16W Vampire Power (Damnjanovic, 2024).

Enclosure

The enclosure stores nearly all the components of the computer. It keeps wires and components protected from most damage. It allows proper airflow so that the computer is protected from dust and overheating.



Figure 8 – Cooler Master Q300L Enclosure

This enclosure (otherwise known as the case) can be positioned horizontal or vertical, and it is designed to have high air performance and extra space for hidden cable management (Amazon, 2026). This enclosure is the ideal enclosure for this build because it is compatible with the motherboard we chose (micro-ATX) and will be great for ventilation and avoiding dust. Dust can be an issue for machines because it can interfere with internal components and can potentially be a fire hazard, so we chose this enclosure because it has a magnetic dust filter on the front, top, and bottom panels to avoid dust accumulation and still allow airflow through the perforated panels (Amazon, 2026).

Table 9 – Specifications and Requirements for the Cooler Master Q300L Enclosure

Specifications	
Manufacturer	Cooler Master
Model	Q300L (MCB-Q300L-KANN-S00)
Type	Mini-Tower
Front Panel Features	2 x USB 3.0 ports 2 x audio jacks LED control button
Requirements	
Mini-Tower	☑
Holds all internal components	☑

Card Reader

This card reader is an external device. It allows you to read, open and transfer data from the computer to the card inserted into the card reader and vice versa (Andrews, Shelton, & Pierce, 2023). Memory cards might be used in digital cameras, tablets, smartphones, MP3 players, digital camcorders, and other portable devices (Andrews, Shelton, & Pierce, 2023).



Figure 9 – SmartQ 5Gbps Card Reader

We chose this card reader because it has fast data transferring speeds and supports most standard memory cards and CompactFlash™. CompactFlash™ (CF) is a type of flash memory card that is usually used to store data and digital photos (Lee, 2023). Lee states that "Compact Flash card is a non-volatile memory technology. This means that unlike RAM, it does not lose its contents when the power is turned off". This is important because it is more durable than an HDD or SSD and does not use moving parts to read and write data. CF cards typically use a file system that is compatible with Windows 11.

Table 10 – Specifications and Requirements for the SmartQ 5Gbps Card Reader

Specifications	
Manufacturer	SmartQ
Size (data transfer rate)	5Gbps
Requirements	
Supports CompactFlash™.	☑

Optical Drive

This optical disk drive (ODD) is external and can be plugged into a USB port. Most optical drives let you burn music, video, or backup discs (Dell Technologies, 2025). It is important to ensure that the type of disc you purchase matches the drive type (DVD+R discs should be used with DVD+R drives, DVD-R discs should be used with DVD-R drives. DVD+R, DVD-R, or DVD+RW (Single/Dual Layer) can be used on computers with DVD+/-RW DL (Dual Layer) drive) (Dell Technologies, 2025). Blu-ray drives are backward compatible with DVD and CD technologies, and DVD drives are backward compatible with CD technologies. Depending on the drive features, an optical drive might be able to read and write to BDs, DVDs, and CDs. A drive that can write to discs is commonly called a burner (Andrews, Shelton, & Pierce, 2023).



Figure 10 – ORIGBELIE BT669 Optical Drive

This optical drive works with Windows 11 and is considered a plug-and-play external drive. Most modern computers do not have a built-in CD/DVD drive, so we chose an external drive because it is a removable option that still has strong error correction, high speed technology, and reading/writing of CD±R/RW, CD-ROM, DVD±R/RW, DVD-RAM, VCD, and SVCD (ORIGBELIE, 2026). This is an external drive that has an optical disk drive (and that is what we were looking for), but it has many other features. You can use this external drive to burn and read CD DVD, use as an external USB and card slot hub, transfer photo and video files from your camera to your computer, use it for peripherals, and for data transfer and system backups (ORIGBELIE, 2026).

Table 11 – Specifications and Requirements for the ORIGBELIE BT669 Optical Drive

Specifications	
Manufacturer	ORIGBELIE
Model	BT669
Speed(s)	5Gbps
Requirements	
Supported Standards	☑
Supports installation of software and creation of backups	☑

Sound (Onboard & Headset)

We have integrated sound, so we do not require an expansion card with sound ports. The headset has noise-cancellation to provide a quiet working environment to the user. Onboard sound (sound ports embedded on a motherboard) can play and record sound and save it in a file (Andrews, Shelton, & Pierce, 2023).



Figure 11 – Grstltd GT-601 Headset

Our motherboard supports onboard audio because it has an audio chipset that is integrated into the motherboard. The integrated audio includes 8 audio channels and has 3 audio ports (newegg, 2023). The headset has noise cancellation, a built-in microphone, and can be jacked into devices (Amazon, 2026).

Table 12 – Specifications and Requirements for the Grstltd GT-601 Headset

Specifications	
Manufacturer	Grstltd
Model	GT-601 (headset) Realtek ALC887 (onboard)
Special Features	Noise Cancellation Built-in microphone
Requirements	
Jacks for mic and speakers	☑
Headset (noise cancellation)	☑
Speakers not included	☑

Keyboard & Mouse

The keyboard and mouse are used to type, navigate, and access features on the computer. These peripherals are essential for the user to operate the machine. The keyboard is ergonomic and spill-resistant, and the mouse is an optical mouse. When you plug a mouse or keyboard into a USB port, Windows should immediately recognize it and install generic drivers (Andrews, Shelton, & Pierce, 2023).

We chose a keyboard and mouse that was a single package because the products were more cost-effective as a bundle, and the environmental handling fee is a single fee instead of two fees. The keyboard is ergonomic so it can reduce strain for the user if they are doing repetitive typing tasks; it is also spill resistant and ideal for an office space. The mouse is an optical mouse, which means that is uses a light-emitting diode (LED), optical sensor and digital signal processor (DSP) in place of a traditional mouse ball and electromechanical transducer; this is important because it operates with a sealed bottom surface that is not impacted by dust and dirt, and it can be used on different surfaces (Barney, 2023). The keyboard and mouse are simple to connect because they plug into USB ports.

Table 13 – Specifications and Requirements for the Logitech MK120 Keyboard

Keyboard Specifications	
Manufacturer	Logitech
Model	MK120/ FBA_LOG920002565
Special Keys/features	Full-size F keys and number pad Curved space bar
Keyboard Requirements	
Ergonomic	☑
Standard PS/2 or USB connector	☑



Figure 12 – Logitech MK120 Keyboard & Mouse Combo

Table 14 – Specifications and Requirements for the Logitech MK120 Mouse

Mouse Specifications	
Manufacturer	Logitech
Model	MK120/ FBA_LOG920002565
Connection type	USB 2.0
Special buttons/features	Ambidextrous with smooth tracking
Mouse Requirements	
PS/2 or USB connector	☑
Optical mouse	☑

Monitor

The primary output device of a computer is the monitor (Andrews, Shelton, & Pierce, 2023). The LCD (liquid crystal display) monitor is also known as a flat-panel monitor (Andrews, Shelton, & Pierce, 2023). It uses LCD technology to produce an image using a liquid crystal material made of large, easily polarized molecules (Andrews, Shelton, & Pierce, 2023). Figure 13 shows the layers of the LCD panel that together create the image (Andrews, Shelton, & Pierce, 2023).

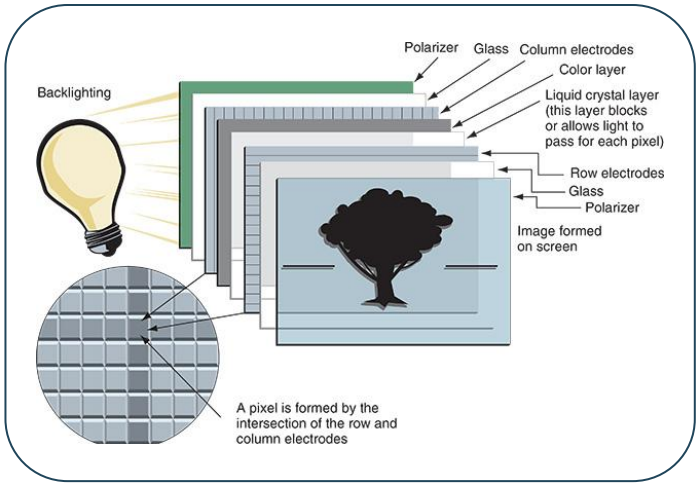


Figure 13 – Layers of the LCD Panel



Figure 14 – ASUS VP227HF Monitor

We chose an LCD monitor with a 22” screen. This full HD (1920 x 1080) frameless monitor is wall-mountable and can save on desktop space (Amazon, 2026) which is ideal for an office space. The client requested a monitor with at least 60Hz but this monitor refreshes content on the display 1.67 times faster than this and has a 100Hz refresh rate (Amazon, 2026). The ASUS DisplayWidget Center is an app that allows the user to tweak monitor settings without needing to use the buttons on the monitor, which a user-friendly feature and helpful in office settings (Amazon, 2026). At Core Systems, we value the environment and sustainability, so we also chose this monitor because it tested to meet the requirements of world-leading sustainability certifications and it is made from materials like 85% post-consumer recycled (PCR) plastics and components like mercury-free panels (Amazon, 2026).

Table 15 – Specifications for the ASUS VP227HF Monitor

Specifications	
Manufacturer	ASUS
Model	VP227HF
Size	22”
LCD/CRT	LCD
Dot pitch	0.249
Maximum resolution	1920 x 1080
Special Features	Adaptive-Sync, HDMI or VGA, 100Hz, 1 millisecond response time, Flicker-free and Low Blue Light technologies

Table 16 – Requirements for the ASUS VP227HF Monitor

Requirements	
Minimum 17”	☑
1920x1080 at 60 Hz or better	☑
Best supports video card’s max res	☑

Conclusion

The proposed plan will be implemented within the allocated timeframe and budget. Every essential requirement that BTB Heavy Equipment outlined in the Request for Proposal (03-2026a) was met and/or exceeded. Every standard desktop computer will be able to run Windows 11 Professional with Office 365 on a Gigabit Ethernet LAN.

Because this proposed plan is under budget, you will be able to make upgrades to these systems or allocate the remaining budget to other business necessities. Each standard desktop computer will cost $933.73[2], and the overall total will be $234,082.50. This saves BTB Heavy Equipment approximately between $15,917.50 - $28,000.00.

Each component was carefully researched and selected to provide BTB Heavy Equipment with the ideal standard desktop computer systems. We made budget allocation decisions such as spending more on a higher performance SSD rather than a traditional hard drive so that the component would provide BTB Heavy Equipment with standard desktop computers that had shorter boot times and higher performance. We chose a power supply that was gold-certified to ensure that the standard desktop computers were unlikely to fail, and in the event of this occurring, the power supply has warranty for the next decade. We were proactive in selecting features for the standard desktop computer to withstand multiple years of use; the machines have USB 3.0 ports and dual channel architecture to support future memory module upgrades. We provided justification and transparency for every decision about the selected components so that BTB Heavy Equipment could make an informed decision about the new standard desktop computers they will implement.

Our team at Core Systems will implement the project plan within 9 weeks; this includes shipping the components, building the new standard desktop computers, data backup, ethical disposal of the current machines, and installation for the new standard desktop computers to the 250 end-users at BTB Heavy Equipment by June 2026.

Ultimately, our skilled team at Core Systems made educated business decisions that will support BTB Heavy Equipment well into the future by selecting components that allow for upgrades and creating these machines while remaining under budget. This is a hands-on practical solution that enhances efficiency, considers existing data and security, focuses on sustainability, and supports your company strategy.

[2] The calculations for this price per unit include environmental handling fees and do not include taxes, labour estimates, liability, or other potential fees. This price does not include components that are on sale at the time of shipping which could make the overall cost decrease.

Budget[3]

Standard Desktop Computer Components				
Part	Notes	Current Sale Cost[4] (per unit)	Individual Cost[5] (per unit)	Bulk Cost (250 units)
CPU			$112.63	$28,175.50
Motherboard			$99.00	$24,750.00
Memory (RAM)			$132.20	$33,050.00
Card Reader			$12.99	$3,247.50
Hard Drive (SSD)			$139.99	$34,997.50
Keyboard & Mouse		$23.74	$24.99	$6,247.50
Optical Drive			$39.99	$9,997.50
Video Card	Integrated	N/A	N/A	N/A
Sound (Onboard & Headset)	Integrated onboard sound		$21.98	$5,495.00
Network Interface Card (NIC)		$24.99	$30.99	$6,247.50
Power Supply			$124.99	$27,497.50
Enclosure			$64.99	$16,247.50
Monitor			$109.00	$27,250.00
Other	FireWire Card		$19.99	$4,997.50
Section Total		$48.73	**$933.73**	**$233,432.50**

Environmental Handling Fees			
Obligated Product for Disposal	Product Description	Environmental Handling Fee (EHF) (per unit)	Environmental Handling Fee (EHF) (250 units)
Desktop Computer	A computer terminal designed to reside on a desk or similar work surface. **May be packaged with a mouse, keyboard, cables, speakers, webcam and other peripherals in a single Original Equipment Manufacturer (OEM) package carrying a single SKU, subject to a single desktop computer EHF.** Includes desktop computers and desktop computers acting as servers	$0.80	$200
Display Devices	A non-portable video display device that typically resides on a table, floor or wall and requires AC power to operate. May contain an imbedded television tuner or may be used for displaying images from computers or other digital or analog sources such as an external TV tuner or cable/satellite receiver. Includes various display technologies, such as CRT, flat panel (LCD, Plasma, LED, OLED, etc.) or rear projection.	**≤ 29": $1.80** 30-45": $3.10 ≥ 46": $7.00	$450
Section Total		$2.60	$650
TOTAL (per unit)		**$936.33**	
TOTAL (250 units)			**$234,082.50**

[3] Please note that taxes, labour estimates, and other potential fees were not calculated due to budget limitations
[4] Sale Cost per unit as of 2026-04-20 (to be subtracted from total per unit if applicable at the time of shipping)
[5] Price per unit as of 2026-04-20

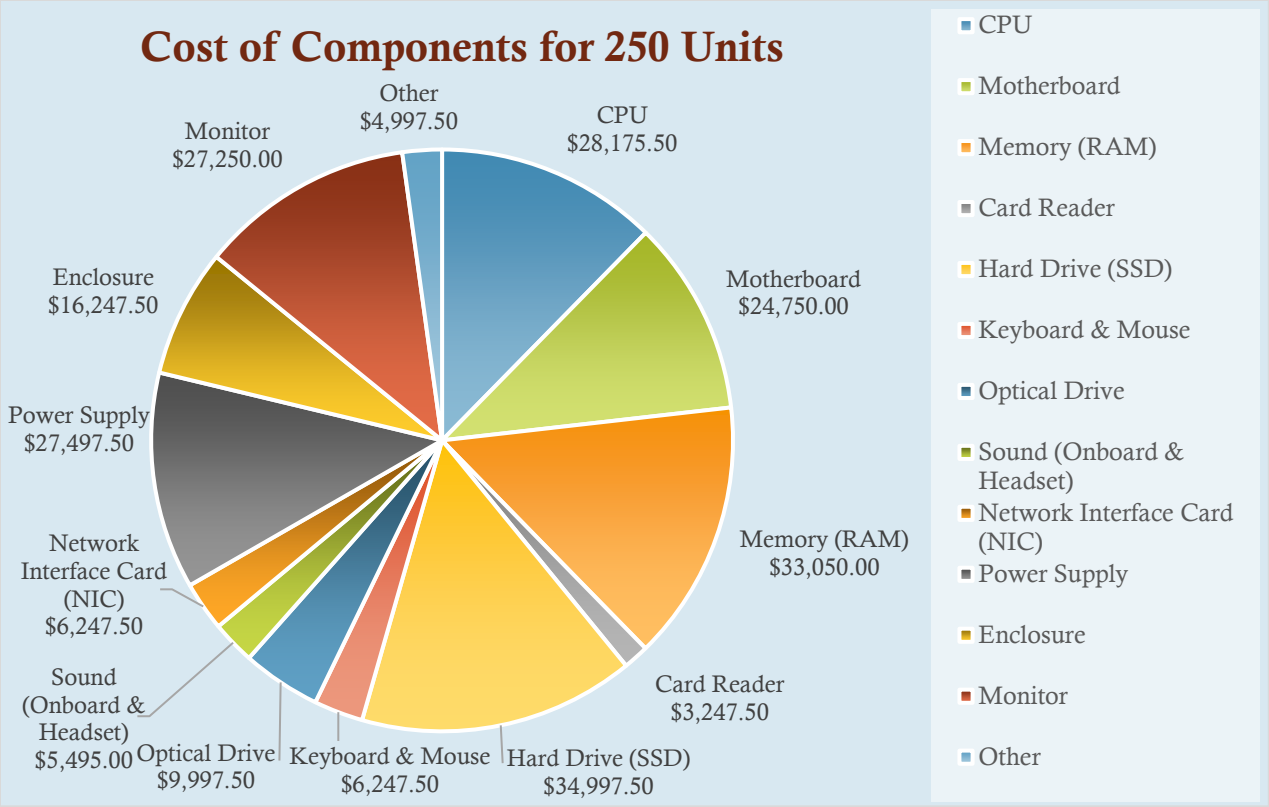


Figure 15 – Breakdown of Cost by Component for 250 Units

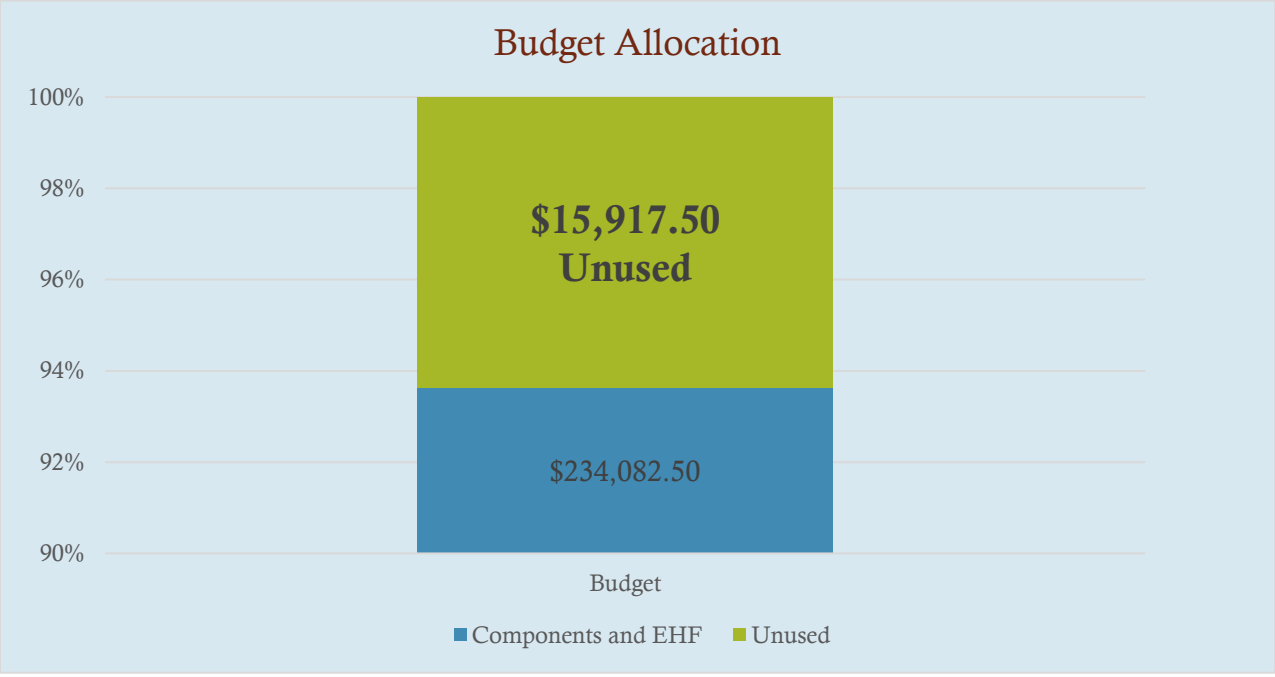


Figure 16 – Budget Allocation

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Appendices

Introduction

This section contains material including but not limited to: The glossary of technical terminology found throughout the proposal, proposed project timeline, disposal (recycling and re-use) solutions, memorandum regarding computer disposal and data backup, RAM considerations, user reviews for the selected components, information about Core Systems (including our team, certifications, and previous successful projects), a copy of the RFP from BTB Heavy Equipment, Inc., and client email communications.

Appendix A – Glossary

AGP — Accelerated Graphics Port; an older expansion interface used for graphics cards before PCI Express became standard.

ATX — A standard size and design format for motherboards and power supplies in desktop computers.

Audio chipset — A hardware component that manages sound processing on the motherboard and handles audio input and output.

Back up data — To make a copy of files so they can be restored if the original data is lost, damaged, or deleted.

Buffered memory — Memory that uses a register between the memory controller and the RAM modules to reduce electrical load and improve stability.

Bus speed — The speed at which data moves between computer components.

Chipset — A group of electronic components on the motherboard that controls communication between the processor, memory, storage, and other hardware.

CompactFlash — A removable flash memory card format used for storing and transferring data in devices such as cameras and card readers.

Connector type — The physical plug or socket standard that allows one device or component to connect to another.

Core — An individual processing unit inside a CPU; multiple cores allow a processor to handle more tasks at the same time.

CPU — Central Processing Unit; the main processor of a computer that executes instructions and controls most operations.

Data integrity — The accuracy, consistency, and reliability of data throughout its lifecycle.

DDR4 — Fourth-generation double data rate memory, a common type of RAM used in modern computers.

Desktop computer — A personal computer designed to stay in one place, usually with separate components such as a monitor, keyboard, and mouse.

DIMM — Dual in-line memory module; a common type of RAM module used in desktop computers and servers.

Discrete graphics card — A separate graphics card with its own graphics processor and memory, unlike integrated graphics.

Dual-channel architecture — A memory configuration that uses two channels to transfer data, improving memory performance.

ECC — Error-correcting code; a memory technology that detects and corrects certain data errors to improve reliability.

Error-correcting code — A method used in memory systems to detect and correct data errors.

Ethernet LAN — A local area network that uses Ethernet cables and equipment for wired communication.

Expansion slot — A slot on the motherboard where add-in cards such as graphics cards or network cards can be installed.

Firmware — Software stored on hardware that controls how a device operates.

Form factor — The physical size, shape, and layout standard of a computer component.

Gigabit Ethernet — A network standard that transfers data at a speed of 1 gigabit per second.

Gigabyte (GB) — A unit of digital storage equal to about one billion bytes.

GPU — Graphics Processing Unit; a processor designed to handle graphics, video, and visual output.

Hard drive — A storage device used to save data permanently, even when the computer is turned off.

IDE — Integrated Drive Electronics; an older interface standard used to connect storage drives.

Integrated component — A part that is built into another device rather than installed separately.

Integrated graphics — Graphics hardware built into the processor or motherboard rather than a separate graphics card.

LAN — Local Area Network; a network that connects computers within a small area such as an office, school, or building.

Latency — The delay before data transfer begins or before a system responds.

LGA-1151 — A CPU socket type used for certain Intel desktop processors.

Micro-ATX — A smaller motherboard form factor than standard ATX, commonly used in compact desktop systems.

Motherboard — The main circuit board in a computer that connects the CPU, memory, storage, and other components.

MTBF — Mean time between failures; the average operating time between one failure and the next in a repairable system.

NIC — Network interface card; hardware that connects a computer to a network.

Noise cancellation — Technology that reduces unwanted background sound, especially in microphones and headsets.

OCP (Over Current Protection) — Over Current Protection or **OCP** has the same purpose, but a different concept as OPP. OCP will generally be faster than OPP, since it uses shunt resistors to check the amount of current, and will shut down if a certain point is reached. OCP on 12v is generally only found on PSUs with multiple rails, since OPP can handle a single rail just fine. It's generally integrated into the protection IC combined with shunt resistors and is a recommended but not required protection by ATX.

Office 365 — Microsoft’s cloud-based productivity suite that includes programs such as Word, Excel, Outlook, and PowerPoint.

Operating system — The main software that manages computer hardware, software, files, and user interaction.

OPP (Over Power Protection) — Over Power Protection or OPP is a protection that will shut down the PSU when too much power is pulled, generally this is between 110 and 140% of the advertised wattage. This is a protection that works as a limit, shutting down when a certain point is reached, but doesn't actively monitor the amount of current. It's generally integrated into the PWM controller and is a crucial protection. It's required in ATX except if OCP is also there.

Optical drive — A drive that reads or writes data on optical discs such as CDs, DVDs, or Blu-ray discs.

OTP (Over Temperature Protection) — Over Temperature Protection or OTP is a protection that protects the PSU against overheating with for example a fan failure. it's generally a thermistor combined with a protection IC that supports this, but there have also been cases where it was integrated into the fan controller. Most reviewers stop measuring after 200c, but it depends on the place the thermistor is integrated what recommended limits are. It's recommended until ATX 2.52, but since has been required.

OVP (Over Voltage Protection) — Over voltage Protection or OVP works to a similar way as UVP, but the other way around. It checks the voltage on a rail that it isn't getting too high, so it can keep a safe height, and will shut down if it gets too high. It's generally integrated into the protection IC and required by ATX standards.

Parity — An error-checking method used in memory systems to help detect corrupted data.

PCI Express — A high-speed connection standard used for expansion cards such as graphics cards and network adapters.

Peripherals — External devices connected to a computer, such as a keyboard, mouse, monitor, or printer.

Power supply unit (PSU) — A component that converts electrical power from an outlet into the correct power needed by computer parts.

Registered DIMM — A type of memory module that uses a register to improve stability, especially in systems that use larger amounts of memory.

Registered memory — Memory that uses a register to reduce electrical load and improve reliability.

RIMM — Rambus inline memory module; an older type of memory module.

SATA III — A storage interface standard used to connect SSDs and hard drives to the motherboard.

SCP (Short Circuit Protection) —Short Circuit Protection or SCP measures the resistance on each rail and will shut down when resistance is lower than 0.1 Ohms. This goes combined with OPP, OCP, OVP and UVP. It's generally integrated into the protection IC, and is required on ATX spec, with separate circuits per rail.

SIMM — Single in-line memory module; an older kind of memory module used in earlier computers.

Socket — The physical connector on a motherboard that holds and connects a CPU.

Socket type — The specific design standard of a CPU socket that determines which processor can fit into a motherboard.

Software installation — The process of placing a software program onto a computer so it can be used.

Solid-state drive (SSD) — A storage device that uses flash memory instead of spinning disks.

Storage device — Hardware used to save digital data, such as an SSD or hard drive.

Throughput — The amount of data that can be processed or transferred in a given amount of time.

USB — Universal Serial Bus; a standard used to connect devices and transfer data and power.

USB 3.0 — A faster version of USB that supports higher data transfer speeds than USB 2.0.

UVP (Under Voltage Protection) — Under Voltage Protection or UVP is meant to shut down if voltage goes down too far. not only because a lot of components need a stable voltage, so you can have less correction on the VRM side, it's also there because OCP and OPP measure by how much current is going through, but not the voltage. If the voltage drops down, the current can go higher, which can result in burnt cables or connectors. It's generally integrated into the protection IC, and recommended in the ATX standard, but required for a modern PSU.

VGA port — A video connector used to send display signals from a computer to a monitor.

Voltage — The electrical force that drives current through a circuit.

Wake on LAN (WOL) — A feature that allows a computer to be turned on remotely through a network connection.

Wired headset — A headset connected by cable that allows audio input and output.

WOL — Abbreviation for Wake on LAN, a feature that allows a computer to be powered on remotely over a network.

Appendix B – Proposed Project Timeline

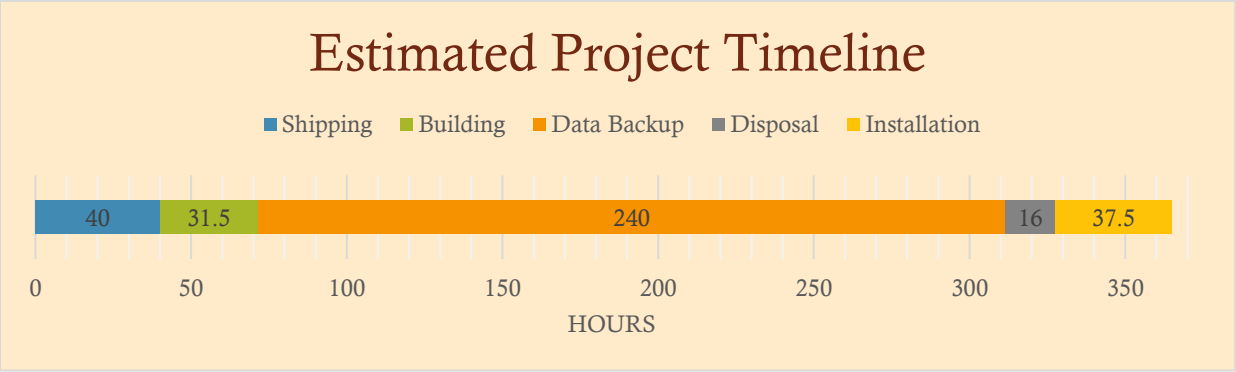
Once selected, Core Systems will order the components for the standard desktop computers and then build each machine at our facility. We have 4 management employees at Core Systems and 16 staff members. We estimate that each build will take approximately 2.5 hours. This would take our team approximately 31.5 hours to complete with 20 team members (this number is calculated from the 625 total hours divided by the number of employees). We estimate that the build alone will take approximately 1 week.

After building the standard desktop computers, Core Systems will deliver them to BTB Heavy Equipment to install. We will back up the data from the old machines, in which the timeline will be dependent on how frequently BTB Heavy Equipment does this and the size of the data to be backed up. It is a recommended best practice for businesses to do this weekly or monthly depending on the data size and required security. If BTB Heavy Equipment has not done this in the past, this could take many hours. For example, a 1TB Full Backup, with a data size of 100GB over 100Mbps bandwidth, would take approximately 22 hours (BestDiskPrices, 2026). This could take up to 6 weeks, but with the available technicians that BTB Heavy Equipment is providing, it will most likely take less time than this estimate.

After the data back-up, we can safely remove the old machines and bring them to our facility for inspection and disposal (see Appendix C for disposal solutions). Then, we will install the new machines at BTB Heavy Equipment and ensure that all the associated drivers are installed, and the new standard desktop computers can run Windows 11 Professional with Office 365 on a Gigabit Ethernet LAN. Installation can take approximately 4.5 hours per computer. We anticipate the installation to take 1,125 hours for all 250 units. With the help of our 20 employees and your 10 technicians, this installation will take approximately 1 week.

In conclusion, we estimate that it will take 365 hours[6] (approximately 9 weeks) to complete this project. This means that we will have to start shipping parts immediately to ensure that the end users receive their standard desktop computers by June of 2026.

Table 17 – Estimated Project Timeline for Each Task in Hours



[6] The budget will not reflect potential labour costs and other associated liability fees

Appendix C – Disposal Solutions (Recycling and Reuse)

There are two options to dispose of the machines. BTB Heavy Equipment can choose to recycle the machines, or they can donate the machines for re-use in the local community.

Recycle the machines:

All recycled machines are subject to the non-refundable environmental handling fee found in the budget. Each machine will be disposed of at the nearest collection site. EPRA, a non-profit association created by electronics manufacturers and retailers, manages the programs that Core Systems collaborates with in accordance with the Province of Saskatchewan's Environmental Management and Protection Act. EPRA is funded by Environmental Handling Fees (EHFs) paid upon the purchase of new electronics (Sarcan Recycling, 2026).

Re-use the machines in the local community:

We value sustainability at Core Systems, so we frequently collaborate with various local community organizations to re-use hardware, peripherals, and software. After we safely back-up data, we take old machines to our warehouse where we determine what is salvageable, and then we donate the machines or components to a local community organization that the client selects. This is the ideal way to dispose of your machine because it will benefit the local community, reduce electronic waste, and make technology accessible for everyone. The client will still be responsible for the non-refundable environmental handling fee found in the budget, but when you donate your components or machine(s), you will receive a text receipt that can be claimed for your business. Organizations qualified for donations include registered charities, non-profit organizations, care facilities, and educational institutions (ERA, 2010).

Our local community partners:

Community Computers Regina Inc.

This Relief of Poverty organization provides free computers to families without home technology. The organization wants everyone in the community to have the chance to succeed in today's digital world. They believe that access to a computer is crucial for education, to search for jobs, and stay connected (Community Computers Regina Inc., 2026).

Computers for Success Saskatchewan

Computers for Success (CFS Saskatchewan) Saskatchewan is delivered by Saskatchewan Technology Renewal Inc. (STRi), a registered charity. CFS Saskatchewan distributes over 4000 computers each year to schools, libraries and non-profits across Saskatchewan (CFS, 2026).

This organization includes Computers for Schools+; CFS+ SK has kept an estimated 19738 pounds of electronic waste out of Saskatchewan landfills through their reuse and recycle program. If it can be reused, they refurbish it so it can support learning, job searching, and digital access in the community. If it can't be reused, they responsibly recycle the parts through approved recycling partners.

The Electronic Recycling Association (ERA)

The ERA is a non-profit organization committed to reducing unnecessary electronic waste by recycling and repurposing used electronics. They work with community charities to provide repurpose computers and electronics to those in need (ERA, 2010).

Appendix D - Memorandum

To: Verna Johnson, CEO of BTB Heavy Equipment, Inc.

From: Skylar Magouliotis, Team Lead at Core Systems

Date: 2026-04-20

Subject: Computer Disposal and Secure Data Backup Procedures

This memo is to inform the staff of BTB Heavy Equipment about the disposal of the current computer systems, and the secure data transfer of all associated data, in compliance with Core Systems and BTB Heavy Equipment's disposal solution (in accordance with the Province of Saskatchewan's Environmental Management and Protection Act) and data protection regulations.

Assets that are scheduled for collection and disposal include:

- Standard desktop computers purchased **before** January 2026
- Associated storage devices (solid-state drives, RAM, optical drives, and any other external drives)
- Peripherals that are considered obsolete or no longer functional

Procedure:

1. All required devices will be wiped of data, backed up, or physically destroyed to ensure no sensitive information remains on the device(s).
2. Hardware will be recycled or reused through the approved electronic waste management facilities in accordance with the Province of Saskatchewan's Environmental Management and Protection Act.
3. Obsolete or unused equipment must be delivered to BTB Heavy Equipment's main building as a collection point for Core Systems to retrieve **by May 11th**.
4. Core Systems will document serial numbers and confirm the disposal for BTB Heavy Equipment's system and machine audit.

If you have equipment that meets these above criteria, please submit it to the head of your IT Services department before the May 11th deadline. Do **not** dispose of, recycle, or attempt to dispose of or recycle any devices independently.

For questions or concerns, please contact Verna Johnson, BTB Heavy Equipment IT Services, or Core Systems.

Skylar Magouliotis

Team Lead
Core Systems

Appendix E – RAM Considerations

The cost of RAM (random access memory) has significantly increased in the last year. The RAM market is experiencing an unprecedented pricing crisis that's reshaping the entire consumer electronics landscape (Ahmmed, 2026).

The companies that manufacture RAM are not prioritizing general users and it is causing an inconsistent and extreme increase in cost. DDR4 (which is what we chose for this project) and DDR5 RAM are both impacted, and this will affect every sector of consumer electronics and enterprise computing (Ahmmed, 2026). This is due to the increased demand of AI (and AI infrastructure) and other factors: the companies that manufacture memory reallocate significant production capacity to HBM (High Bandwidth Memory) for AI data centers and GPUs, and this reduces consumer RAM supply; scarcity is created when there is a deliberate phase-out of DDR4 production in favour of DDR5 production; there is consumer demand for higher memory capacities (driven by AI PC's and local AI models); and production cuts from past oversupply contribute to an industry that is unable to quickly respond to increased demand (Ahmmed, 2026).

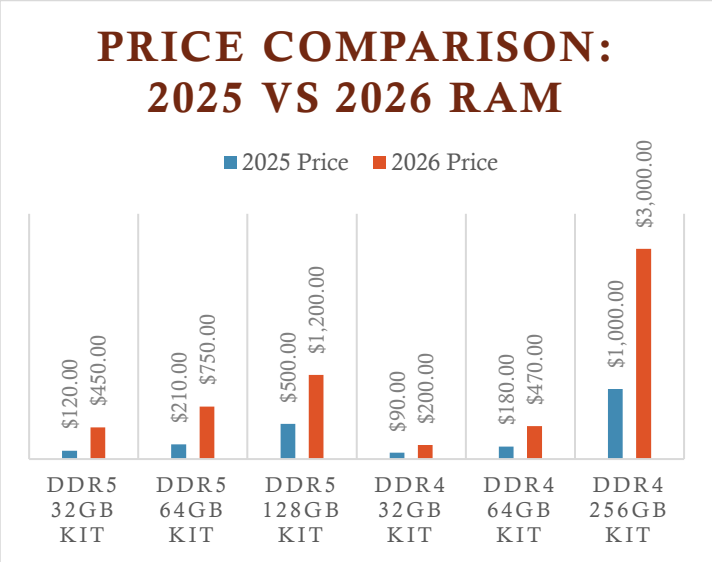


Figure 17 – Price Comparison (highest recorded price) of RAM in 2025 vs 2026

We recommend that businesses phase projects into stages, implement baseline configurations now, and expand later when conditions improve (Ahmmed, 2026). It is BTB Heavy Equipment's best interest to start with adequate (but not excessive) RAM capacity and anticipate expansion when prices normalize in the next 2-3 years. That is why we selected 8GB of DDR4 RAM for every machine; it is the minimum recommended requirement to run Windows 11 Professional with Office 365 and will operate efficiently on a Gigabit Ethernet LAN. Our team will recommend an additional 8GB of RAM when the client can support this cost, but adding an extra RAM module to 250 units is estimated to cost an additional $34,000.00 which is not within scope for this project at this time.

Table 18 & Figure 17 demonstrate the severity of the change in RAM prices between 2025 and 2026.

Table 18 – RAM prices in 2025 vs 2026 and the change in %

Memory Type	**2025 Price**	**2026 Price**	**Change (%)**
DDR5 32GB Kit	$90-120	$310-450	+300-400%
DDR5 64GB Kit	$180-210	$710-750	+294-357%
DDR5 128GB Kit	$400-500	$1,000-1,200	+140-200%
DDR4 32GB Kit	$60-90	$150-200	+150-233%
DDR4 64GB Kit	$120-180	$350-470	+161-292%
DDR4 256GB Kit	$800-1,000	$3,000+	+200-275%

Appendix F – User Reviews for Selected Components

User Reviews (for selected components)

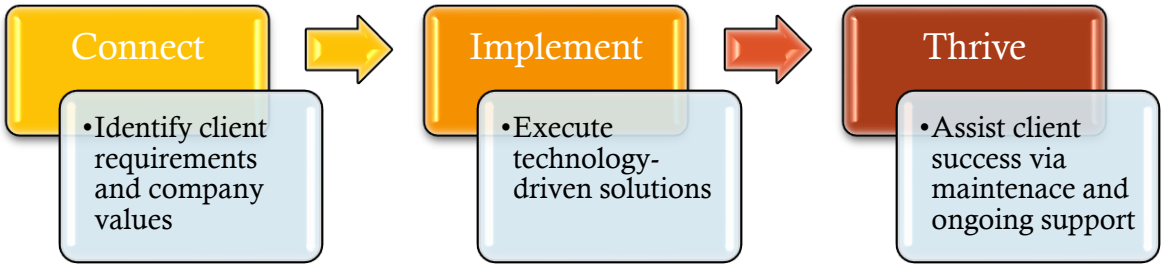
Component	Review #1	Review #2
Central Processing Unit (CPU)	"This is a major step up from the 2nd gen core i3 computer that it is replacing. Installed in an ASUS TUF B360-PLUS Gaming motherboard. It appears to be a fast processor, and [the number of] cores are a plus." (Estabrook, 2019).	"I do recommend this CPU for budget gamer. Or for a pc build of [a low-end price] range." (Anonymous, does a job, does it well, 2019).
Motherboard	"I have had no issues out of it since installation. The BIOS has tons of options and easy to navigate. Overall, a great board for the price." (A, 2016).	"For a basic board, this little guy is very nice. The BIOS was easy to use and I'm actually a little jealous of it over the one in my main system from a different manufacturer. It was a set it and forget it deployment. No strange issues, no crashes." (Anonymous, Rock Solid, 2016).
Card Reader	"Worked great. Amazing how a product as reasonably priced as this, can do such a great job." (V.K., 2026).	"This little gadget seems to work well, no matter what I've thrown at it. It's USB 3.0, so it works much faster than the older 2.0 versions. I would definitely recommend if you're looking for an affordable way of being able to work with literally any kind of memory card, including the older ones like MemoryStick Pro for older digital cameras." (Lamothe, 2026).
Hard Drive (SSD)	"My Lenovo 17" laptop had slowed through the years to the point that it took almost 5 minutes to fully boot. I changed out the 1TB 2.5" HD Seagate HD for this SSD. The laptop now boots in less than 30 seconds. The SSD worked so well that I doubled the memory from 8GB to 16GB. Now this old beast runs like a new one. I couldn't be happier." (Cary, 2022).	"The integrity of these drives is remarkable. They stand the test of time and dependability! You can pay more for someone else's name but this is the best Bang for your Buck!" (Wayne, 2022).
Keyboard & Mouse	"I've spent over 25 years developing software… The keyboard's tactile feedback is sublime, making typing a joy. The keys are well-spaced, and the layout is intuitive. The mouse, too, is a revelation. Its optical sensor tracks smoothly, and the ergonomic design fits perfectly in my hand." (Bruce, 2025).	"Just can't go wrong with this keyboard and mouse. Nice and quiet for when your mechanical keyboard strains the relationship with others within earshot (the main reason for this purchase)." (D, 2015).
Optical Drive	"True plug-and-play No drivers needed—just connect via USB-A or USB-C and it works right away on most systems. USB 3.0 provides solid transfer speeds. It's not blazing fast, but perfectly fine for installing software, watching DVDs and ripping or burning CDs. The built-in 8-in-1 hub is also a big plus." (Kreamer, 2026).	N/A
Sound (Headset)	"I recently started using the Grstltd Wired Headset with Mic for work-from-home meetings and virtual classes, and it's honestly been a great upgrade from my	"Wired, over the ear, lightweight, boom mic. It is difficult to find all of these specs in a headset that has a good price and sounds good both in my ears and on the other end

	previous generic earbuds. The headset is plug-and-play with the standard 3.5mm jack, so it works perfectly with both my laptop and tablet without any extra setup. The sound quality is crisp and clear for both listening and speaking" (Tomoya, 2025).	to my customers. These work great." (Branscum, 2026).
Network Interface Card (NIC)	"Works well in Windows and Linux. No latency or connection dropping issues. OS was able to use this right away without installing vendor drivers." (Y, 2023).	"It works. No issues. I went from a 1gb to 2gb internet. I installed the card, plugged into the 2.5gb port on my router and it showed 2ish gb. I would buy it again." (W, 2023)
Power Supply	"Easy and very clean installation. I love its modularity. No unused cables blocking my airflow. More than enough hardware and wires. Super quiet. Cute bag to store unused wires." (Anonymous, Great Product, 2021).	"It's quiet has the option to switch the fan mode It was a good price for power It's compact for an 850-watt unit. With pc parts costing more and everyone talking about tier list, I don't overclock. I've never had a problem with power supplies, so this will do fine for midrange builds." (S, 2025).
Enclosure	"Purchased this case for a budget Micro-ATX build. The case is very well built. Lots of value for your money here. It comes with everything you need, an exhaust fan, screws and even a bag of zip ties! Cable management was a breeze, plenty of space and holes to route wires through" (Phillippe, 2023).	"For this price the case is perfect. I'd love to see a tempered glass panel but again for this price I can't complain. Fit my micro atx motherboard. Looks great and cooling is actually quite easy inside this case" (IV, 2026).
Monitor	"I have been using this monitor for 2 months with no complaints." (Anonymous, 2026).	"Very good monitor a little smaller than I thought ... So now I'm going to buy a bigger and keep this as a back up" (Anonymous, Easy Hook Up <<, 2026)
FireWire Card	"Immediately Recognized (Win 11). Needed to hook up an old digital video camera to archive the tapes. Installed board, powered up computer, connected Firewire from the camcorder to the card and it was immediately recognized... no hassles or configuration problems (using video capture software.) Great prices and works great!" (Thielking, 2024).	"I did some research and found this PCIe Card. The only concern I have was that I only have a 16 pin PCIe slot available as my 1 pin was covered by my graphics card, but it was no issue once I learned it was fine and compatible. It was a smooth "plug and play". I have a Win 11 System and I really didn't have to download any drivers after installing it. The computer loaded normally, didn't crash or anything, which is a plus." (Emily, 2026).

Appendix G – About Core Systems

Core Systems
Connect, Implement, Thrive

Core Systems is a local Saskatchewan-based IT company. We offer consultations, create new systems, and maintain existing systems. We deliver reliable, efficient, and cost-effective technology services to businesses. We provide technology solutions to clients so that they can operate efficiently in a modern digital economy. We work closely with our clients to identify their specific requirements, implement technology that improves workflow, and offer future support to guarantee that the client is successful.



The specialized team at Core Systems is trained and certified to help your business thrive. Our team offers consultations, custom computer builds, deployment, data management and security, and ethical disposal solutions. We build machines based on client requirements, provide supervision and additional technicians for machine installations, and assist with after-sale support for machine and system maintenance. We are skilled in data transfer and backup, and we ensure data integrity and security. We value environmental sustainability, and we work with organizations to provide community support and accessibility as well as ethical disposal in accordance with the Province of Saskatchewan's Environmental Management and Protection Act as described in Appendix C.

Core Systems will deliver practical and hands-on solutions for all your IT concerns through our technical knowledge and commitment to quality service. Our team is dedicated; we provide systems and machines, protect data, and support businesses throughout Saskatchewan.

Our Core Values

Trust

Passion

Sustainability

Teamwork

Innovation

Honesty

Collaboration

Respect

Integrity

Commitment

Quality

Community

Core Systems – Meet our Management Team

Core Systems consists of 4 management employees and 16 staff members. Team members are all cross-trained and highly skilled.



Skylar Magouliotis

Team Lead

Project Management
Operational Efficiency & Consulting
Data Analysis



Savneet Kaur

Software Engineer

Programming
Database Management
AI



Matthew Livingstone

Master Builder & Installation Technician

Hardware
UX Design
Troubleshooting



Lucas McCulloch

Network Administrator & Security Analyst

Cloud Computing
Cybersecurity
Data Transfer

Our specialized team holds widely recognized certifications: the CompTIA A+ Certification, which is the industry standard credential and covers hardware configuration, software installation, preventative maintenance, networking, security, and more; the CompTIA Network+ Certification, which covers network technologies, installation and configuration, media and topologies, management, and security; the Microsoft Certified Solutions Expert (MCSE) Certifications (this includes Cloud Platform and Infrastructure, Mobility, and Data Management and Analytics certifications) which validate expertise in Windows Server, cloud computing, platform virtualization, deploying and managing Windows devices and applications, and covers skills in data management, analysis, business intelligence, and big data analytics using Microsoft solutions; the Cisco Certified Network Associate (CCNA) certification, which covers skills like installation, configuration, administration, security, and troubleshooting of Cisco networking devices; and the Apple Certified Mac Technician (ACMT) certification, which shows we have mastered the intricacies of Mac hardware repair. In addition to these certifications, our team has WHMIS training and can handle electrical and dangerous materials. Our team are also Microsoft 365 Certified Administrator Experts and can deploy, implement, and manage Microsoft 365.

CERTIFICATIONS

CompTIA A+ ✦ CompTIA Network+ ✦ MSCE ✦ CCNA
ACMT ✦ WHMIS ✦ Microsoft 365

Core Systems – Previous Successful Projects

Dinosaur Sports

Dinosaur Sports is a local charity that provides every child with the opportunity to play their desired sport. Core Systems created and implemented a database that kept track of the children registered for each sport, and which leagues had availability for more children to register. We designed, tested, and implemented the system within 3 weeks with no issues.

> *"The staff at Core Systems was so friendly and considerate. They seemed to really care about what our organization does. They were able to do this project for us faster than we anticipated and were honest about the budget and timeline. Another company told us this would take 2 months, but Core Systems gave their expect advice and they were able to do it faster which saved us time and money!"*
> *- Janice Betellum, CEO of Dinosaur Sports*

Just Videos

Just Videos is a local Saskatchewan VFX (visual effects) company that specializes in documentaries about the Plains Cree Indigenous people. Core Systems designed and built high-quality computers that were able to handle the strain of VFX such as computer-generated imagery, compositing, motion capture, and match moving. We built 40 intricate computer systems within a budget of $50,000.00 last year (2025).

> *"Their team was extremely fast and very professional. They did thorough research before working with us to determine what machines would work best for our needs within our budget. They installed the machines for us, and they answered all our questions with honesty and good advice. I will be recommending them to everyone I know!"*
> *- Jane Debbie Pearson, Operational Manager of Just Videos*

Smash Building

Smash Building is a demolition company that reached out to Core Systems from word of mouth. They were impressed with the services we had provided to other local businesses and asked if we could do a consultation. Core Systems built them 100 computers that could meet requirements for drone systems and 3D mapping that they use for demolitions. We designed and built 100 of these machines with a budget of $100,000.00 and completed this project in March of 2026 within their timeline.

> *"When my friend from Just Videos mentioned the fast service and deployment of 40 machines for an excellent price, I knew I had to reach out to Core Systems immediately. Their team was highly skilled and pleasant to work with, and they ended up saving us $10,000!"*
> *– Jesse Frankman, CEO of Smash Building*

Appendix H – RFP from BTB Heavy Equipment, Inc.

BTB Heavy Equipment, Inc.



PO Box 1520
Saskatoon, SK S7K 3R5
(306) 659 – 4256
(306) 659 – 4267

Request for Proposal (03-2026a)

BTB Heavy Equipment (BTB) was formed in 1999 and contracts out highly automated and customized cranes, bulldozers, and other equipment to construction firms across the country. BTB requires new standard desktop computers, which will be delivered to about 250 end users in June of 2026. The proposal must describe, in detail, the new machines and their component parts. The firm that wins the contract will source and build the new machines and will supervise BTB's computer systems technicians in their installation.

About BTB: offering contracts for the most reliable brands with exceptional customer service, BTB is here to satisfy your heavy equipment needs. We make it easy for you to get your work done and are committed to deliver the results you need. We are committed to building a trusted relationship with you and supporting Saskatchewan businesses.

Submitted proposals must have the following perimeters:

- Not exceed the capital budget of $250 000 Cdn[1]
- Describe computers to be built from parts, not bought as an entire package from one vendor
- Consist of machines able to run Windows 11 Professional with Office 365 on a Gigabit Ethernet LAN

Appendix I – Client Email Correspondence

Outlook

Re: RFP Preliminary Questions and Insights - Core Systems Team

From Johnson, Verna <johnson6225@saskpolytech.ca>
Date Fri 2026-04-10 12:12 PM
To Magouliotis, Skylar <magouliotis4413@saskpolytech.ca>

Hi Skylar,

Thank you for reaching out! Here are the answers to your questions:

In order to achieve this, we had some preliminary questions:

1. What does BTB stand for? - **The original founders chose the name BTB Heavy Equipment. BTB are the initials of the founders grandmother. You can just refer to us as BTB Heavy Equipment Inc.**

2. Time efficiency is important to our team at Core Systems; How many technicians are available for installation at BTB Heavy Equipment, Inc? - **There will be as many technicians as you need. This won't affect the cost in your proposal.**

3. Data security is a core value at Core Systems. Are your technicians also responsible for backing up important data and wiping previous data from your current hard drives? If they are not responsible for this, our firm offers protection against data loss and we can offer you this service. - **We'd appreciate your team providing your recommendations on how to back up, wipe, and dispose of (or donate) the old machines. If your team also provides this service, we'd be happy to leave it in your hands.**

4. Core Systems is a local Saskatchewan business and we are always interested in community initiatives. If the current computers are in good condition, would your company be interested in the re-use of these computers or do you wish to recycle them? Our team at Core Systems believes that the community programs through local libraries and schools can benefit from using these computer systems if they are not completely end-of-life yet. - **We'd be more than happy to donate or recycle the old machines.**

4. Core Systems is a local Saskatchewan business and we are always interested in community initiatives. If the current computers are in good condition, would your company be interested in the re-use of these computers or do you wish to recycle them? Our team at Core Systems believes that the community programs through local libraries and schools can benefit from using these computer systems if they are not completely end-of-life yet. - **We'd be more than happy to donate or recycle the old machines.**

5. Our team at Core Systems considers the environmental impact of the technology that we build and implement. We would want to ensure that the current computers are being safely and ethically disposed of, therefore we will need to know the following information:
    a. What is the size(s) of the current monitors? - **24"**

    b. Are your current systems all-in-one systems or were they packaged as a single OEM package with a single SKU? If not, are your current keyboard and mouse peripherals originally packaged in one OEM box or are they separate products? - **The computers came as complete packages with these peripherals.**

    c. Are there any other peripherals or parts that are not listed in the RFP that will need to be disposed of? - **No.**

6. Is there any important information our team should know about BTB Heavy Equipment Inc. that isn't explicitly stated in the Request for Proposal? - **No, that should cover it!**

Feel free to reach out if you have any other questions.

Verna Johnson
CEO
BTB Heavy Equipment Inc.

From: Magouliotis, Skylar <magouliotis4413@saskpolytech.ca>
Sent: April 9, 2026 8:48 PM
To: Johnson, Verna <johnson6225@saskpolytech.ca>
Subject: RFP Preliminary Questions and Insights - Core Systems Team

Dear Mrs. Johnson,

Our team at Core Systems received a Request for Proposal from BTB Heavy Equipment, Inc. and we are interested in addressing all of your needs.

It is essential for your business to make an informed decision about our partnership. Core Systems is committed to building a trusted relationship with every one of our partners and supporting Saskatchewan Businesses.
The team at Core Systems is interested in gaining more insight into your company to ensure that we can deliver the results you need, and experience alignment on both of our goals and values.

In order to achieve this, we had some preliminary questions:

1. What does BTB stand for?

2. Time efficiency is important to our team at Core Systems; How many technicians are available for installation at BTB Heavy Equipment, Inc?

3. Data security is a core value at Core Systems. Are your technicians also responsible for backing up important data and wiping previous data from your current hard drives? If they are not responsible for this, our firm offers protection against data loss and we can offer you this service.

4. Core Systems is a local Saskatchewan business and we are always interested in community initiatives. If the current computers are in good condition, would your company be interested in the re-use of these computers or do you wish to recycle them? Our team at Core Systems believes that the community programs through local libraries and schools can benefit from using these computer systems if they are not completely end-of-life yet.

5. Our team at Core Systems considers the environmental impact of the technology that we build and implement. We would want to ensure that the current computers are being safely and ethically disposed of, therefore we will need to know the following information:
    a. What is the size(s) of the current monitors?
    b. Are your current systems all-in-one systems or were they packaged as a single OEM package with a single SKU? If not, are your current keyboard and mouse peripherals originally packaged in one OEM box or are they separate products?
    c. Are there any other peripherals or parts that are not listed in the RFP that will need to be disposed of?
6. Is there any important information our team should know about BTB Heavy Equipment Inc. that isn't explicitly stated in the Request for Proposal?

With this insight, Core Systems can ensure that we are able to deliver the results that BTB Heavy Equipment Inc. requested and guarantee that this partnership is successful for both of our businesses.

We look forward to your response.

Thank you,
Skylar Magouliotis
Team Lead

Core Systems